

POSLink Android AIDL Guide

Overall Description

This document describes how to use the communication type AIDL on POSLink. In some of Android POS devices, we have Android applications called BroadPOS that handles transactions as a standalone POS machines. And BroadPOS allows other applications integrated POSLink SDK to connect them to do transactions. The connection protocol bases on AIDL(Android Interface Definition Language).

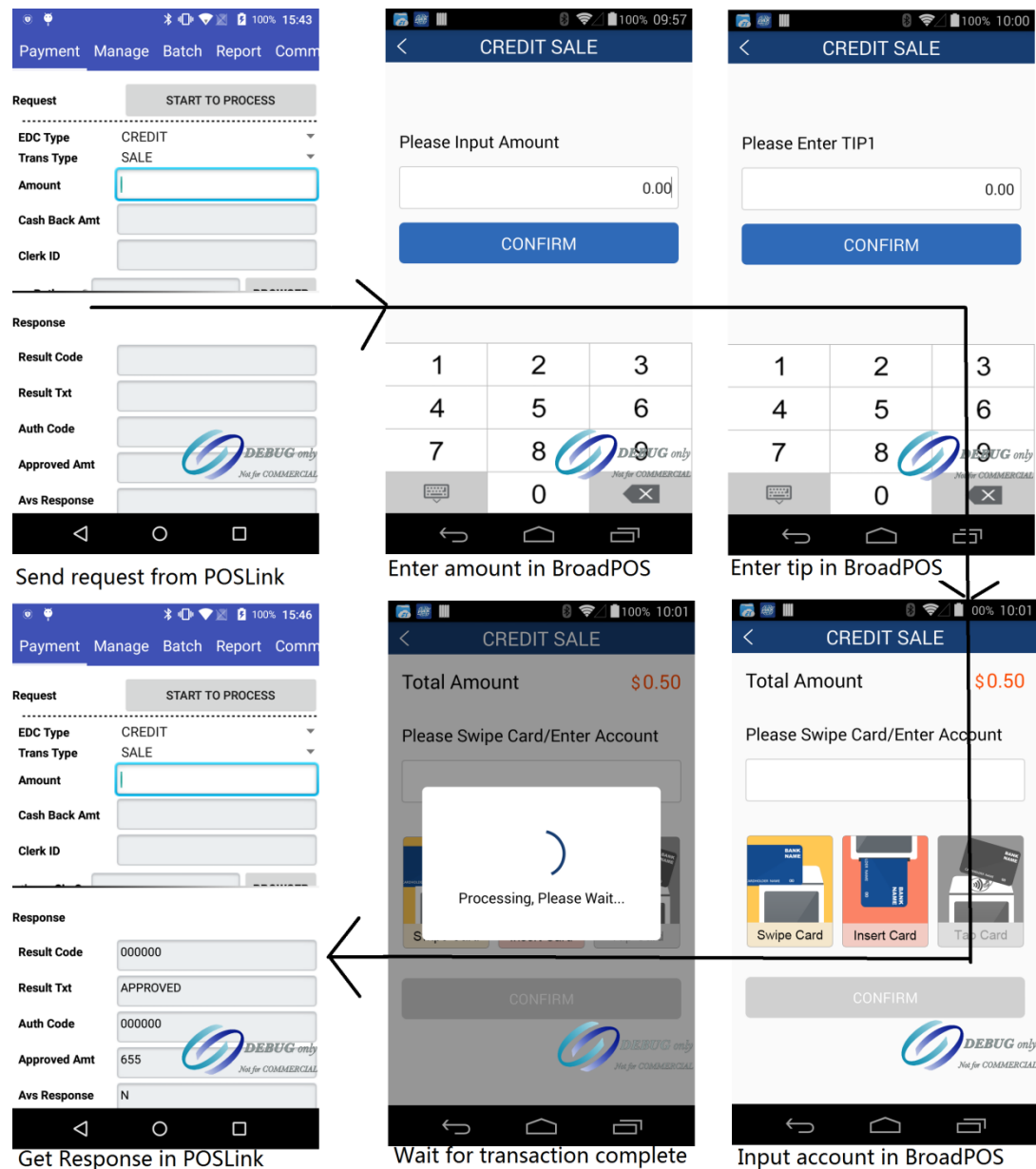
Product Features

This type of communication is quite different from other communication as it is an interprocess communication. And it can only be used in Android devices.

The main feature of AIDL is that when you start POSLink, your application will jump to BroadPOS. And then you can input amount, swipe card and connect to transaction host to finish the transaction. The POSLink still accept some fields like amount before jumping to BroadPOS. And after the transaction, response will be sent to your application.

The supported features are quite different from old POSLink. So this document will tells what POSLink AIDL supports now.

Simple Sample



targetSdkVersion requirements

```
<!-- AIDL, targetSdkVersion greater than or equal to 30 -->
<queries>
    <intent>
        <action android:name="com.pax.us.std.poslink.aidl" />
    </intent>
</queries>
```

Simple Code Guidance

Use `CommSetting#setType(CommSetting#AIDL)`. Then use `POSLink#ProcessTrans()` to do the transactions.

Supported Transactions

Payment

Request

Value	Attribute	Description
TenderType *		ALL = 0 –U CREDIT = 1 DEBIT = 2 CHECK = 3 EBT_FOODSTAMP = 4 EBT_CASHBENEFIT = 5 GIFT = 6 LOYALTY = 7 CASH = 8 EBT = 9 Setting any other value will generate an error response.
TransType *		See PaymentTransType below Setting any other value will generate an error response.
Amount	n...8	Transaction total amount\$\$\$\$\$CC.
CashBackAmt	n...8	Cash back amount \$\$\$\$\$CC.
FuelAmt	n..8	Fuel Amount \$\$\$\$\$CC. It included in transaction amount.
ClerkID	ans...8	Employee/Clerk ID.
Zip	n...9	Cardholder/Check Owner billing zip/postal code.
TipAmt	n...8	Tip amount \$\$\$\$\$CC.
TaxAmt	n...8	Tax amount \$\$\$\$\$CC. It included in transaction amount.
Street	ans...32	Billing street address.
Street2	ans...32	Backup billing street address.
ServerID –U		
SurchargeAmt	n...8	Merchant surcharge fee, \$CC. If the terminal level setting for “merchant fee” is “0”, this field must be NULL or “0”. If the terminal level setting is not “0”, this field can be any value or “NULL”, if the value is “NULL”, terminal will fall back to the default value. Supported only in Debit sale and EBT cash benefit sale transactions.
PONum	ans...32	Purchase order number.
OrigRefNum	n...4	Original terminal reference number used for follow-up

		transactions.	
InvNum	ans...16	POS system invoice/tracking number. Its max length is 16, but its type and length are host depended.	
ECRRefNum *	ans...16	The ECR reference number.This is a unique code on theECR side.	
AuthCode	ans...10	Voice Authorization Code obtained via telephone from the payment host. The transaction type is “FORCED”. This field is NULL and the terminal will prompt for the authorization.	
ExtData	<ExpDate>	n4	Card expire date, MMY format
	<TableNum>	n...4	Table Number
	<GuestNum>	n...4	Guest Number
	<HRef>	ans...32	Host reference number (or Transaction UID). This field is host dependent; it can be used to run Void/Return transactions. If this field is returned, suggest to print in the receipt, ans..32.
	<CustomerEnrollmen tCode>	ans...64	The unique number to identify a customer
	<Token>	ans..64	Token number, ans..64

Response

Response	Attribute	Value Description
ResultCode *	ans6	Transaction Result Code
ResultTxt	ans...32	Transaction Result Txt Review result txt for transaction details.
AuthCode	ans...10	Returns the transaction authorization code from the payment processor. Suggest printing this on the receipt.
ApprovedAmount	n...8	Approved Amount Displays the actual amount approved by the host. This

		could be different from the requested amount.	
AvsResponse	ans...8	AVS response Displays the AVS response.	
BogusAccountNum	n...19	The number returned to the POS. Two possible conditions: Semi-Integration: Displays the last 4. Full Integration: Displays the full account number.	
CardType	var	Refer to <u>card type</u> definition, it presets card type name. Card type is determined by the bin range.	
CvResponse	ans...8	CVV response code.	
HostCode	ans...32	Host Reference Number Suggest printing this on the receipt.	
HostResponse	ans...8	Host Response or Error Code	
Message	ans...32	Host or Gateway Message	
RefNum	n...4	Reference Number/Transaction Number 0 < Index <= 5000, normal data base. 5000 < Index <= 9000, stored data base for SAF. 9000 < Index <= 9999, failed data base for SAF.	
RemainingBalance	n...8	If the EDC is GIFT The card holder's balance, \$\$\$\$\$CC. If the EDC is EBT, this is the cash benefits balance. Or LOYALTY or CREDIT, this is host returned balance.	
ExtraBalance	n...8	The card holder's balance, \$\$\$\$\$CC. If the EDC is EBT, this is the food stamp balance. If the EDC is LOYALTY, this field will be the host returned additional balance.	
RequestedAmount	n...8	Requested Transaction Amount	
Timestamp	n14	Transaction Time/Date Stamp	
ExtData	<PLCardPresent>	n1	Display Whether or Not the card Was Present at the Time of the Transaction Valid value: 0: Card Present 1: Card Not Present Mandatory for manually card entry. Credit transactions only.

	<PLEntryMode>	n1	Card Entry Type/Method Valid value: 0:Manual 1:Swipe 2:Contactless 3:Scanner 4.Chip 5.Chip Fall Back Swipe
	<PLNameOnCard>	ans..32	Cardholder Name
	<PLPONumber>	ans..32	Transaction PO Number
	<PLCustCode>	ans...32	Customer Code
	Tags below applicable for amount information.		
	<AmountDue>	n...8	Amount Due:

			(\$\$\$\$\$\$CC) in a partial authorization
	<TipAmount>	n...8	Tip Amount: (\$\$\$\$\$\$CC). Mandatory for tips
	<CashBackAmount>	n...8	Cash back amount \$\$\$\$\$CC.
	<MerchantFee>	n...8	Merchant Fee Amount: (\$\$CC). Mandatory for merchant fee
	<TaxAmount>	n...8	Tax Amount: (\$\$\$\$\$\$CC). Mandatory for tax amount
Tags below applicable for account information.			
	<ExpDate>	n4	Card Expiration Date: (MMYY). Mandatory for manually entry
	<EBTFoodStampVoucher>	a1	Food Stamp Voucher

			Valid only when tender type =EBT_FOODSTAMP or EBT. T: True F: False
	<VoucherNum>	ans...16	Voucher Number Mandatory when the EDC = EBT or EBT type = Voucher.
	<CVVMessage>	ans...32	CVV response message Mandatory if host returns a CVV message. Credit transaction only.
Tags below applicable for trace information.			
	<ECRRefNum>	an...16	ECR Reference Number (Echo back).
	<AVSMessgae>	ans...32	AVS Response Message

			Mandatory if the host returns an AVS response.
Additional Information			
<TableNum>	n...4		Table Number
<GuestNum>	n...4		Guest Number
<FPSOffline>	n1		FPS Offline Transaction indicator, 1 = Transaction amount is below the required Authorization threshold. Transaction approved offline.
<FPSSignStatusNum>	n1		FPS Offline Transaction indicator, 1 = Transaction amount is below the required Signature threshold. No signature required
<FPSReceiptModeNum>	n1		FPS Receipt Mode 1 = Merchant copy only 2 = Customer copy only

			3 = No receipt Default = Print Both
	<ORIGTIP>	n...8	The original tip amount for CREDIT ADJUSTMENT
	<Token>	ans...64	Token value return from the processor
	<HRef>	ans...32	Host reference number (or Transaction UID). This field is host dependent; it can be used to run Void/Return transactions. If this field is returned, suggest to print in the receipt, ans..32.
	<CARDBIN>	n...6	The card BIN as the first 6 digits of account number. or If the transaction type is REPLACE, this field is the old card number

	<SN>	ans..32	The serial number of the device
	Tags below applicable for EMV Contact/Contactless trans only.		
	<TC>	ans..16	EMV Transaction Certificate, tag 9F26
	<TVR>	ans...32	EMV Terminal Verification Results, tag 95
	<AID>	ans..10	EMV Application Dedicated File (ADF) Name Application ID, tag 4F
	<TSI>	ans..32	EMV Transaction Status Information, tag 9B.
	<ATC>	ans..4	EMV Application Transaction Counter, tag 9F36.
	<APPLAB>	ans..4	EMV Application Label, tag 50.

	<APPPN>	ans..16	EMV Application Preferred Name, tag 12
	<IAD>	ans..16	Issuer Application Data, tag 9F10.
	<ARC>	ans..64	AuthorizationResponse Code, tag 8A. Y1 – Offline approved Y3 – Unable to go online, offline approved Z1 – Offline declined Z3 – Unable to go online, offline declined Other – Online approved or declined.
	<CID>	ans..2	Cryptogram Information Data, tag 9F27
	<CVM>	n1	Cardholder Verification Method (CVM) Codes, tag 9F34 0 – Fail CVM processing 1 - Plaintext Offline

			PIN Verification 2 – Online PIN 3 – Plaintext Offline PIN and Signature 4 – Enciphered Offline PIN Verification 5 – Enciphered Offline PIN Verification and Signature 6 – Signature 7 – No CVM Required.
Tags below applicable for FAILED EMV.			
	<AC>	ans..6	Contact/Contactless trans only. Authorization Code, tag 89
	<AIP>	ans..4	Application Interchange Profile, tag 82
	<AVN>	ans..4	Application Version Number, tag 9F08

	<IAUTHD>	ans..32	Issuer Authentication Data, tag 91
	<CDOL2>	ans..256	Card Risk Management Data Object List 2 (CDOL2), tag 8D
	<HRED>	ans..256	Host Responded EMV Data, may include tag 8A/89/91/71/72
	<IAD>	ans..16	Issuer Application Data, tag 9F10.
	<ARC>	ans..64	AuthorizationResponse Code, tag 8A. Y1 – Offline approved Y3 – Unable to go online, offline approved Z1 – Offline declined Z3 – Unable to go online, offline declined Other – Online approved or declined.
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	<AC>	ans..6	Contact/Contactless trans only. Authorization Code, tag 89

	<AIP>	ans..4	Application Interchange Profile, tag 82
	<AVN>	ans..4	Application Version Number, tag 9F08
	<IAUTHD>	ans..32	Issuer Authentication Data, tag 91
	<CDOL2>	ans..256	Card Risk Management Data Object List 2 (CDOL2), tag 8D
	<HRED>	ans..256	Host Responded EMV Data, may include tag 8A/89/91/71/72

Batch

Request

Value	Attribute	Description
TransType *		BATCHCLOSE = 1 – Closes the current batch.

Response

Response	Attribute	Value Description	Remarks
ResultCode *	ans6	Transaction Result Code: Determines transaction results.	
ResultTxt	ans...32	Transaction Result Text	Review transaction

			details.
CreditCount	var	Total Credit Transaction Count	Does not include voids or authorizations.
CreditAmount	var	Total Credit Transaction Amount	Does not include voids or authorizations
DebitCount	var	Total Debit Transaction Count	
DebitAmount	var	Total Debit Transaction Amount	
EBTCount	var	Total EBT Transaction Count	
EBTAmount	var	Total EBT Transaction Amount	
GiftCount	var	Total Gift Transaction Count	
GiftAmount	var	Total Gift Transaction Amount	
LoyaltyCount	var	Total Loyalty Transaction Count	
LoyaltyAmount	var	Total Loyalty Transaction Amount	
CashCount	var	Cash Transaction Count	
CashAmount	var	Total Cash Transaction Amount	
CHECKCount	var	Total CHECK Transaction Count	
CHECKAmount	var	Total CHECK Transaction Amount	
Timestamp	n14	Transaction Time/Date Stamp: Date/Time format: YYYYMMDDhhmmss	
TID	ans...20	Terminal ID	
MID	ans...20	Merchant ID	
HostTraceNum	ans...32	Host Trace Number: Mandatory when returned by the host. Suggest to print this on the receipt.	
BatchNum	ans...32	Batch Number: Mandatory when returned by host.	
AuthCode	ans...10	Authorization Code: Returns the transaction auth code from the payment processor.	
HostCode	ans...32	Reference Number: Payment processing host reference number.	
HostResponse	ans...8	Payment processing host response	
Message	ans...32	Host or Gateway Message	

Manage

Request

Value	Attribute	Description
TransType *		GETVAR = 2 –This command is used to get the control variable and configure parameter in the terminal. In POSLink dll version 1.00.25 or higher, this command support to get multi-variables' value. If all variables' value get successful, it will be return ok, or error will be returned with indicated which one is error. If one VAR get failed, all other VARs will fail. SETVAR = 3 –This command is used to set the control variable and configure parameter in the terminal. In POSLink dll version 1.00.25 or higher, this command support to set multi-variables' value. If all variables' value set successful, it will be return ok, or error will be returned with indicated which one is error. . If one VAR set failed, all other VARs will fail.

Response

Response	Attribute	Value Description	Remarks
ResultCode *	ans6	Transaction Result Code: Determines transaction results.	Result code of transaction. Used to determine the results of a transaction.
ResultTxt	ans...32	Transaction Result Text	Review result txt for details of a transaction.
VarValue	ans...60	The Value of the Variable	Valid when the requested transaction =GETVAR
VarValue1	ans...60	The Value of the Variable 1	Valid when the requested transaction =GETVAR
VarValue2	ans...60	The Value of the Variable 2	Valid when the requested transaction =GETVAR
VarValue3	ans...60	The Value of the Variable 3	Valid when the requested transaction =GETVAR
VarValue4	ans...60	The Value of the Variable 4	Valid when the requested transaction =GETVAR

